

Higher-Order Bounded Sign: Four Positive Regimes and an Abstract No-Go

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Status. Expanded partial result, likely valid and requiring human review. The source question is answered affirmatively for fixed symbol range, for semidefinite D , when the compressed coefficient is self-adjoint, and whenever the active compression is a one-sided spectral multiplier perturbation $I + P_+ M_b P_-$. This last theorem allows rotating range, both spectral signs, and non-self-adjoint coefficient. An abstract counterexample shows why the full direction-dependent case cannot follow from accretivity alone.

Source question and framework

Auscher, Axelsson, and McIntosh consider $\mathcal{H} = L^2(\mathbb{R}^n; \mathbb{C}^m)$, a homogeneous constant-coefficient differential operator D of order k , and a bounded multiplication operator B [1]. Their hypotheses (H1)–(H5) say that D is self-adjoint,

$$\|\nabla^k u\|_2 \lesssim \|Du\|_2 \quad (u \in \mathcal{D}(D) \cap \overline{\mathcal{R}(D)}),$$

and

$$\operatorname{Re}(BDu, Du) \geq \delta \|Du\|_2^2.$$

On p. 2, immediately after Corollary 1.3, they ask whether for $k > 1$ these hypotheses and the multiplication structure of B already imply that $\operatorname{sgn}(BD)$ is bounded. Their Theorem 1.1 proves the required quadratic estimate after adding off-diagonal hypothesis (H6), and Proposition 1.2 deduces boundedness of $\operatorname{sgn}(BD)$ once that estimate is known for B and B^* .

Idea of the proof

The obstruction in order $k > 1$ is that a cutoff commutator $[D, \eta]$ contains all derivative orders below k . If D has dense range, however, (H3) is a full elliptic estimate. The resolvent equation controls $t^k Du$, ellipticity controls $t^k \nabla^k u$, and Fourier interpolation controls every $t^j \nabla^j u$. Thus conjugating the resolvent by an exponential cutoff changes it by a uniformly small operator from the scaled graph domain to L^2 . A Neumann series gives exponential off-diagonal decay, precisely (H6).

When the symbol range is a fixed subspace V , the kernel is the local fixed space $V^\perp$. Compressing B to V preserves multiplication and accretivity. The dense-range argument applies on $L^2(V)$, and elementary similarities transfer the sign calculus back to BD .

For arbitrary symbol range, the same algebraic compression still works but is nonlocal. It immediately settles two further regimes: a positive compression is similar to a self-adjoint operator when it is itself self-adjoint, while a semidefinite D leaves only one spectral half-plane. Finally, an explicit bilateral discrete-Hilbert-transform example makes the sign unbounded for an abstract

accretive compression with exponentially spaced spectrum. Thus a full proof must exploit Euclidean polynomial growth and multiplier locality.

That polynomial growth can in fact be exploited for every one-sided spectral compression $I + P_+ M_b P_-$. The spectral graph satisfies a linear Sylvester equation. Its Fourier kernel is split according to the relative sizes of the input and output frequencies: comparable scales are Coifman–Meyer, the high-output region is a standard high–low paraproduct, and the low-output region gains a summable power of the scale ratio. The resulting $L^\infty \times L^2 \rightarrow L^2$ bound gives an exact similarity to the positive/negative diagonal operator. The remaining general case is harder precisely because two-sided coupling replaces this linear equation by a nonlinear Riccati equation.

are involved (Carleson measures, $T(b)$ argument). Furthermore, our proof allows a simultaneous treatment of higher order D , i.e. $k \geq 2$, which is new.

The interest of proving a quadratic estimate is mainly in the following proposition as a corollary of results developed in [22].

Proposition 1.2. *Assume that B, D satisfy (H2,4,5) on a Hilbert space $\mathcal{H}$, that BD satisfies the quadratic estimate (1), and that B^*D satisfies the same quadratic estimate with B^* in place of B . Then the operator $\text{sgn}(BD)$ is bounded on $\mathcal{H}$ and invertible on $\overline{R(BD)}$.*

The operator $\text{sgn}(BD)$ is zero on $\mathbf{N}(BD)$ and satisfies $(BDBD)^{1/2} = \text{sgn}(BD)BD$ on $\mathbf{D}(D)$. More is true, in particular BD has a bounded holomorphic functional calculus on $\mathcal{H}$. We remark that the specific nature of $\mathcal{H}$, B and D is not used in this proposition, which follows from operator theoretic considerations, once quadratic estimates for the operators BD and B^*D have been proved.

When $k = 1$, we obtain the following corollary to this result, once we have proved Proposition 5.1. Note that if B satisfies (H4,5), then so does B^* .

Corollary 1.3. *Assume that B, D satisfy (H1-5) on $L^2(\mathbf{R}^n, \mathbf{C}^m)$, that $k = 1$, and that B is multiplication by a function $B \in L^\infty(\mathbf{R}^n, \mathcal{L}(\mathbf{C}^m))$. Then the operator $\text{sgn}(BD)$ is bounded on $\mathcal{H}$ and invertible on $\overline{R(BD)}$.*

When $k > 1$, we do not know if B being a multiplication operator is enough, in addition to (H1-5), to conclude for the boundedness of $\text{sgn}(BD)$. It is the case when B, D are as in Section 2.1 and the boundedness of $\text{sgn}(BD)$ appears new.

Known consequences of the boundedness of operators $\text{sgn}(BD)$ are short proofs of the Kato conjecture for elliptic systems in divergence form [4, 5] and the boundedness

The source proposition, first-order result, and higher-order question (locally rendered arXiv:0810.3071v2, p. 2).

Dense-range off-diagonal lemma

Lemma 1. *Assume (H1)–(H5), let B be a multiplication operator, and suppose $\overline{R(D)} = \mathcal{H}$. Then, for both choices of sign,*

$$R_t^\pm = (I \pm it^k BD)^{-1}$$

has exponential L^2 off-diagonal decay at scale t . Consequently (H6) holds for B, D , the same is true for B^, D , and $\text{sgn}(BD)$ is bounded.*

Proof. Density extends (H5) to every $v \in \mathcal{H}$. Hence B is strictly accretive on $\mathcal{H}$ and has a bounded inverse. The source’s bisectorial resolvent estimate under (H2), (H4), and (H5) gives

$$\|u\|_2 \lesssim \|(I \pm it^k BD)u\|_2.$$

Writing $L_t^\pm = I \pm it^k BD$ and using B^{-1} gives

$$t^k \|Du\|_2 \lesssim \|L_t^\pm u\|_2.$$

Since the range is dense, (H3) applies to every $u \in \mathcal{D}(D)$. Homogeneity and the corresponding upper estimate show that D is elliptic, $\mathcal{D}(D) = H^k$, and

$$t^k \|\nabla^k u\|_2 \lesssim \|L_t^\pm u\|_2.$$

The elementary Fourier inequality $(t|\xi|)^j \lesssim 1 + (t|\xi|)^k$, $0 \leq j \leq k$, now yields the uniform scaled graph estimate

$$\sum_{j=0}^k t^j \|\nabla^j u\|_2 \lesssim \|L_t^\pm u\|_2. \quad (1)$$

Let closed sets E, F have distance $d > 0$ and first suppose $t \leq d$. Choose a smooth ϕ equal to 1 on E and 0 on F , with $\|\nabla^j \phi\|_\infty \lesssim d^{-j}$ for $1 \leq j \leq k$. Put $g = \alpha d \phi / t$, where $0 < \alpha \leq 1$ will be fixed. Leibniz' rule and $t \leq d$ give

$$e^g D e^{-g} = D + K_g, \quad t^k \|K_g u\|_2 \lesssim \alpha \sum_{j=0}^{k-1} t^j \|\nabla^j u\|_2.$$

Indeed, a term in which j derivatives hit e^{-g} has coefficient at most $C\alpha t^{-j}$; products of several derivatives of g only add powers of $\alpha \leq 1$. Combining this with (1) shows

$$\|it^k B K_g (L_t^\pm)^{-1}\|_{2 \rightarrow 2} \leq C\alpha.$$

Choose α so that $C\alpha < 1/2$. Then

$$e^g L_t^\pm e^{-g} = L_t^\pm \pm it^k B K_g$$

is invertible with a uniform L^2 inverse by a Neumann series. If f is supported in F , then $e^g f = f$, whereas $e^g = e^{\alpha d/t}$ on E . Therefore

$$\|1_E R_t^\pm f\|_2 \lesssim e^{-\alpha d/t} \|f\|_2.$$

For $t > d$ the uniform resolvent bound gives the same estimate after changing the constant. Finally,

$$t^k B D (I + t^{2k} B D B D)^{-1} = \frac{1}{2i} (R_t^- - R_t^+),$$

so (H6) follows, with exponential rather than merely polynomial decay. The same proof applies to B^* because it has the same accretivity real part. Theorem 1.1 and Proposition 1.2 of [1] now imply boundedness of $\text{sgn}(BD)$. $\square$

Active-space reduction

Let P be the orthogonal projection onto $\mathcal{R} = \overline{\mathcal{R}(D)}$, let $\mathcal{K} = \mathcal{N}(D)$, and write D_R for the restriction of D to $\mathcal{R}$.

Lemma 2. *Put $C = PB|_{\mathcal{R}}$ and $U = B|_{\mathcal{R}}$. Then C is boundedly invertible on $\mathcal{R}$, U is an isomorphism from $\mathcal{R}$ onto $B\mathcal{R}$, and*

$$\mathcal{H} = B\mathcal{R} \dot{+} \mathcal{K}, \quad B D U = U D_R C.$$

Consequently boundedness of $\text{sgn}(BD)$ is equivalent to boundedness of $\text{sgn}(D_R C)$.

Proof. By (H5) and density of $\mathcal{R}(D)$ in $\mathcal{R}$, $\text{Re}(Cv, v) \geq \delta \|v\|^2$ on $\mathcal{R}$, so C is invertible and U is bounded below. For $h \in \mathcal{H}$, set $v = C^{-1}Ph$. Then $h - Bv \in \mathcal{K}$, and the sum is direct because $PBv = Cv$. Finally, $DBv = DPBv = D_R Cv$, which gives the intertwining identity. Functional calculus is invariant under the displayed bounded similarity and is zero on $\mathcal{K}$. $\square$

Three affirmative regimes

Let $\widehat{D}(\xi)$ denote the homogeneous matrix symbol of D .

Theorem 1. *Assume (H1)–(H5), let B be a bounded multiplication operator, and suppose there is a fixed subspace $V \subseteq \mathbb{C}^m$ such that*

$$R\widehat{D}(\xi) = V \quad (\xi \neq 0).$$

Then $\text{sgn}(BD)$ is bounded on $L^2(\mathbb{R}^n; \mathbb{C}^m)$ and invertible on $\overline{R(BD)}$.

Proof. Here P is the constant pointwise projection onto V , so $C = PB|_V$ is again a multiplication operator. The operator D_R has dense range on $L^2(V)$, and Lemma 1 gives bounded $\text{sgn}(CD_R)$. Since $D_R C = C^{-1}(CD_R)C$, Lemma 2 transfers the bounded sign to BD . $\square$

Theorem 2. *Under (H1)–(H5), $\text{sgn}(BD)$ is bounded in either of the following cases:*

- (i) *D is nonnegative or nonpositive;*
- (ii) *$C = PB|_{\mathcal{R}}$ is self-adjoint on $\mathcal{R}$; in particular, this holds when B is self-adjoint on $\mathcal{H}$.*

No fixed-range hypothesis is required.

Proof. If $D_R \geq 0$, then CD_R is sectorial in the right half-plane, by strict accretivity of C , and its sign on $\mathcal{R}$ is I . If $D_R \leq 0$, its sign is $-I$. Since $D_R C = C^{-1}(CD_R)C$, conclusion (i) follows from Lemma 2.

For (ii), (H5) gives $C \geq \delta I$ on $\mathcal{R}$. The operator

$$A = C^{1/2} D_R C^{1/2}, \quad D(A) = C^{-1/2} D(D_R),$$

is self-adjoint, and $D_R C = C^{-1/2} A C^{1/2}$. Thus $\text{sgn}(D_R C)$ is bounded, and the reduction again finishes the proof. $\square$

A general one-sided spectral-triangular theorem

Let P_+ and P_- be the positive and negative spectral projections of D_R , and write $\mathcal{R}_{\pm} = P_{\pm} \mathcal{R}$. Hypothesis (H3) gives a uniform gap between zero and the nonzero eigenvalues of the homogeneous symbol. Thus $P_{\pm}$ are smooth degree-zero matrix multipliers away from the origin, and

$$D_R = \begin{bmatrix} A_+ & 0 \\ 0 & -A_- \end{bmatrix} \quad \text{on } \mathcal{R}_+ \oplus \mathcal{R}_-,$$

where $A_{\pm}$ are positive bundle multipliers whose eigenvalues are comparable to $|\xi|^k$.

Lemma 3 (Bundle Sylvester estimate). *For $b \in L^{\infty}(\mathbb{R}^n; \mathcal{L}(\mathbb{C}^m))$, set*

$$N = P_+ M_b P_- : \mathcal{R}_- \longrightarrow \mathcal{R}_+.$$

There is a bounded $X : \mathcal{R}_- \rightarrow \mathcal{R}_+$ such that

$$A_+ X + X A_- = 2A_+ N, \quad \|X\| \lesssim_D \|b\|_{\infty}. \quad (2)$$

Proof. Let $a_+(\zeta)$ and $a_-(\eta)$ denote the positive bundle symbols. For a matrix H , let $\mathbf{m}(\zeta, \eta)[H]$ be the unique bundle map x solving

$$a_+(\zeta)x + xa_-(\eta) = 2a_+(\zeta)p_+(\zeta)Hp_-(\eta). \quad (3)$$

The positive Sylvester map on the left has inverse norm at most $C_D/(|\zeta|^k + |\eta|^k)$. Define X by the bilinear Fourier multiplier whose kernel is (3), with $H = \widehat{b}(\zeta - \eta)$. We prove its endpoint bound by a smooth partition into three frequency-ratio regions. The separation constant is chosen small enough that the two Neumann series below converge in every finite collection of multiplier seminorms; polynomial factors produced by differentiating the j th term are absorbed by its geometric ratio gain.

When $|\zeta| \simeq |\eta|$, the normalized bundle symbol is smooth and homogeneous of degree zero. After factoring off the bounded input and output multipliers P_- and P_+ , it is an ordinary bilinear Coifman–Meyer symbol. The standard endpoint estimate $L^\infty \times L^2 \rightarrow L^2$ therefore applies [2, Introduction].

Suppose next that $|\eta| \ll |\zeta|$ and put $K = p_+(\zeta)Hp_-(\eta)$. Multiplying (3) on the left by $a_+(\zeta)^{-1}$ gives the norm-convergent expansion

$$x = 2 \sum_{j \geq 0} (-1)^j a_+(\zeta)^{-j} K a_-(\eta)^j. \quad (4)$$

The $j = 0$ term is the standard high–low paraproduct; its $BMO \times L^2 \rightarrow L^2$ bound applies because $L^\infty \subset BMO$. On each dyadic ratio layer, the j th remaining term gains $(|\eta|/|\zeta|)^{kj}$. The normalized angular multipliers are uniformly Calderón–Zygmund, so the paraproduct estimates and the geometric series in j and in the ratio layers give the required bound.

Finally, when $|\zeta| \ll |\eta|$, multiplication on the right by $a_-(\eta)^{-1}$ gives

$$x = 2 \sum_{j \geq 0} (-1)^j a_+(\zeta)^{j+1} K a_-(\eta)^{-j-1}. \quad (5)$$

Every term now gains $(|\zeta|/|\eta|)^{k(j+1)}$. On the layer $|\zeta| \simeq 2^{-\ell}|\eta|$, the output projection of the product is bounded by

$$C_D 2^{-k\ell} \|b\|_\infty \|v_\eta\|_2.$$

Littlewood–Paley orthogonality sums the input scales and the geometric ℓ -sum closes. Equations (4) and (5), together with the comparable region, prove $\|X\| \lesssim_D \|b\|_\infty$. Equation (2) follows from (3) in distributions. $\square$

Theorem 3. *Assume (H1)–(H5). If, relative to $\mathcal{R} = \mathcal{R}_+ \oplus \mathcal{R}_-$, the active compression has the form*

$$C = PB|_{\mathcal{R}} = \begin{bmatrix} I & P_+ M_b P_- \\ 0 & I \end{bmatrix}$$

for some bounded matrix function b , then $\text{sgn}(BD)$ is bounded. The same holds with the lower-triangular orientation reversed.

Proof. Let X be given by Lemma 3 and put

$$R = \begin{bmatrix} I & -X/2 \\ 0 & I \end{bmatrix}, \quad J = \begin{bmatrix} A_+ & 0 \\ 0 & -A_- \end{bmatrix}.$$

The Sylvester identity says $A_+(N - X/2) = XA_-/2$. Hence $N - X/2$ maps $D(A_-)$ into $D(A_+)$, and a direct calculation, including the reverse domain inclusion, gives the closed-operator identity

$$D_R C = R J R^{-1}.$$

Functional calculus under this bounded similarity yields

$$\operatorname{sgn}(D_R C) = R \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix} R^{-1} = \begin{bmatrix} I & X \\ 0 & -I \end{bmatrix}.$$

Lemma 2 transfers the bounded sign to BD . Taking adjoints, or reversing the two bundles, proves the lower-triangular case. $\square$

For example, Theorem 3 contains the following genuinely rotating-range family in every even order. On $L^2(\mathbb{R}^n; \mathbb{C} \oplus \mathbb{C}^n)$ let

$$\widehat{D}_r(\xi) = \begin{bmatrix} |\xi|^{2r} & 0 \\ 0 & -|\xi|^{2r-2} \xi \xi^* \end{bmatrix}, \quad B_\beta(x) = \begin{bmatrix} 1 & \beta(x)\ell \\ 0 & I_n \end{bmatrix}.$$

Here the negative range is the radial line and the kernel is the tangential plane. If $\|\beta\|_\infty \|\ell\| < 2$, then B_β is strictly accretive and the theorem applies. In this case the Sylvester symbol reduces to the explicit Coifman–Meyer function

$$\frac{2|\alpha + \eta|^{2r}}{|\alpha + \eta|^{2r} + |\eta|^{2r}}.$$

Why accretivity alone is insufficient

The next example is not a Euclidean differential counterexample. It shows that the locality and polynomial spectral growth in the source problem are indispensable.

Theorem 4 (Abstract no-go). *There are an injective self-adjoint operator D_0 with dense range and a bounded uniformly strictly accretive operator C_0 on a Hilbert space for which $\operatorname{sgn}(C_0 D_0)$ is unbounded.*

Proof. Let $H_0 = \ell^2(\mathbb{Z})$, fix $r > 1$, and set $Ae_j = r^j e_j$. Let T be bilateral convolution with $h_d = 1/d$ for $d \neq 0$ and $h_0 = 0$. Its Fourier symbol is the bounded sawtooth series $\sum_{d \neq 0} e^{id\theta}/d$, so T is bounded. On $H_0 \oplus H_0$ put

$$D_0 = \begin{bmatrix} A & 0 \\ 0 & -A \end{bmatrix}, \quad C_0 = \begin{bmatrix} I & \varepsilon T \\ 0 & I \end{bmatrix}, \quad 0 < \varepsilon \|T\| < 1.$$

The Hermitian part of C_0 is bounded below by $1 - \varepsilon \|T\|/2 > 0$.

If $\operatorname{sgn}(C_0 D_0)$ were bounded, triangular functional calculus would give

$$\operatorname{sgn}(C_0 D_0) = \begin{bmatrix} I & X \\ 0 & -I \end{bmatrix}, \quad AX + XA = -2\varepsilon TA.$$

Matrix entries force, for $i \neq j$,

$$X_{ij} = \frac{-2\varepsilon}{(i-j)(1+r^{i-j})}.$$

Thus X would be convolution by $x_d = -2\varepsilon/[d(1+r^d)]$. For $d \rightarrow -\infty$ this has the one-sided harmonic tail $-2\varepsilon/d$, while for $d \rightarrow +\infty$ it decays exponentially. Its Fourier series is therefore a nonzero multiple of $\log(1 - e^{-i\theta})$ plus a bounded function, and is not essentially bounded. Such convolution is unbounded on $\ell^2(\mathbb{Z})$, a contradiction. $\square$

Consequences and remaining obstruction

The hypothesis of Theorem 1 holds whenever D is elliptic ($V = \mathbb{C}^m$), whenever D is scalar and nonzero under (H3), and more generally whenever the kernel of $\widehat{D}(\xi)$ is independent of $\xi \neq 0$.

Theorem 2 also settles every semidefinite system and every self-adjoint coefficient, even when the symbol range rotates. Theorem 3 goes further for every homogeneous symbol D : it settles all one-sided spectral compressions $I + P_+ M_b P_-$ and their lower-triangular analogues, including non-self-adjoint examples with both signs and rotating range. The unresolved core therefore requires genuinely two-sided coupling between the positive and negative spectral bundles; having both signs, rotating range, and non-self-adjoint compression is no longer by itself an obstruction.

This is not merely a missing estimate in the proof above. If $R\widehat{D}(\xi)$ varies with direction, the projection onto $\overline{R(D)}$ is a nonlocal homogeneous Fourier multiplier. Compressing B to that range therefore destroys the local multiplication structure, and exponential conjugation of the projection is not uniformly bounded. Theorem 4 shows that accretivity alone cannot replace this lost locality. A full solution must exploit polynomial homogeneous spectrum and the fact that B is multiplication, or construct a Euclidean example that defeats those structures. For a general block compression, eliminating one off-diagonal block introduces inverses of nonlocal Toeplitz-type compressions; equivalently the linear Sylvester equation in Theorem 3 becomes a nonlinear Riccati equation without a contraction parameter.

Verification and novelty check

The graph estimate tracks every power of t and derivative order in the higher-order commutator; its constant is independent of t, d, E, F . The argument was checked separately for B^* , as required by source Proposition 1.2. The active-space reduction was checked without assuming that the range projection is local. In the no-go example, the forced off-diagonal block is a convolution operator and its unbounded logarithmic Fourier symbol gives an exact contradiction, not numerical evidence.

For Lemma 3, the audit separates all three frequency-ratio regions. Comparable scales use the cited endpoint Coifman–Meyer estimate; the high-output region is the standard $BMO \times L^2$ paraproduct plus a geometric Sylvester series; and the low-output region has the explicit $2^{-k\ell}$ gain needed to sum its ratio layers. The fibrewise Sylvester identity then gives the operator identity in distributions. Finally, $A_+(N - X/2) = XA_-/2$ supplies the exact domain mapping in the triangular similarity. The displayed even-order family independently checks the formula against its scalar explicit symbol.

No numerical assertion enters the proof. Finite Fourier experiments were discarded because they produced the same apparent growth in the known first-order case. The run’s four cheap indexes had no hit for the source id or theorem. Bounded local citation scans and web searches on 9 August 2026 for the exact question, higher-order BD calculus, dense range, fixed kernel, and fixed symbol range found the source and adjacent first-order/structured Kato work. Additional exact-phrase searches for triangular coefficients, spectral compressions $P_+ M_b P_-$, rotating-range signs, and the Sylvester/Coifman–Meyer mechanism found the source and general paraproduct literature, but no explicit statement of Theorem 3. Novelty confidence is moderate because the literature check was bounded and the area is specialized.

Human-review recommendation. Check the Leibniz bound for K_g in Lemma 1, the unbounded one-sided harmonic Fourier series in Theorem 4, and the domain statement for $C^{1/2} D_R C^{1/2}$ in Theorem 2. For the new theorem, verify the three-region dyadic estimate in Lemma 3, with particular attention to the BMO paraproduct at $j = 0$, and the domain passage in the triangular similarity.

References

- [1] P. Auscher, A. Axelsson, and A. McIntosh, *On a quadratic estimate related to the Kato conjecture and boundary value problems*, Harmonic Analysis and Partial Differential Equations, Contemp. Math. **505** (2010), 105–129; arXiv:0810.3071v2.
- [2] S. Arias and S. Rodríguez-López, *Some endpoint estimates for bilinear Coifman–Meyer multipliers*, arXiv:2005.11771v1 (2020).